

EXPERIMENT – 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

AIM: To study GPIO control and bit-addressable SFR using Embedded C.

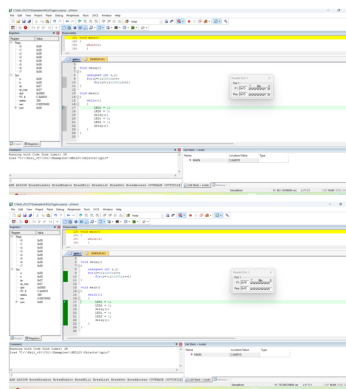
OBJECTIVES:

1. To understand GPIO operation of the AT89C51.
2. To access individual port bits using sbit.
3. To toggle P1.0 and P1.1 alternately.

PROGRAM:

```
#include <reg51.h>
sbit LED1 = P1^0;
sbit LED2 = P1^1;
void delay()
{
    unsigned int i,j;
    for(i=0;i<200;i++)
        for(j=0;j<1000;j++);
}
void main()
{
    while(1)
    {
        LED1 = 1;
        LED2 = 0;
        delay();
        LED1 = 0;
        LED2 = 1;
        delay();
    }
}
```

OUTPUT IMAGE:



OBSERVATION:

Register	Value
P1	0x01
P1	0x02

RESULT:

Thus GPIO and bit addressing was studied and in Port1 the pins P1.0 and P1.1 toggled alternately.

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